

Scalability & Future Plan

Date	6 July 2026
Team ID	
Project name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum marks	1 mark

Scalability & Future Plan

The OptiCrop system is designed to support future growth by handling more users, larger agricultural datasets, and additional smart farming features. The application can be enhanced with advanced technologies to improve prediction accuracy and provide better support to farmers.

Current System Limitations

S.No	Limitation	Impact	Priority
1.	Limited agricultural dataset	Reduces prediction accuracy	High
2.	No real-time weather integration	Affects recommendation quality	High
3.	Supports limited crop varieties	Limits usability	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1.	Uer Load	Supports limited users	Deploy on cloud for multiple users

2.	Data Storage	Local database	Migrate to cloud database
3.	Performance	Basic ML prediction	Optimize ML model and caching
4.	Security	Basic authentication	Add secure login and data encryption

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Real-time weather integration	3 Months	Better crop prediction
Phase 2	IoT sensor integration	6 Months	Live field monitoring
Phase 3	Mobile application	9 Months	Easy farmer acces
Phase 4	Multi-language support	12 Months	Wider adoption by farmers